

14.01 — Principles of Microeconomics

125 points

Problem Set 1 — Graded

Fall 2023

Student: Amin Cheipournejad · Total: 93 / 125

Grade Report

Part	Score	Grader's comment
Q1 1.1	2 / 2	Positive, with the correct justification — the statement reports a measurable quantity.
Q1 1.2	2 / 2	Normative. Correct.
Q1 1.3	2 / 2	Positive. Correctly recognises that a causal claim is testable, and therefore positive, regardless of whether it is true.
Q1 1.4	2 / 2	Positive. Correct.
Q1 1.5	0 / 2	Marked positive; the answer is normative. The test applied throughout the section — whether the statement says how things should be — is too narrow to catch an evaluative claim phrased as a description. “Good for the people of the United States” is a value judgement that no measurement settles.
Question 1 subtotal	8 / 10	

Part	Score	Grader's comment
Q2 2.1	4 / 5	Utility function $x + y$ correct, with two straight downward-sloping curves drawn. Deduction: axes unlabelled and the direction of increasing utility not indicated, both of which the question requires.
Q2 2.2	3.5 / 5	The right-angled curve is correct for perfect complements and the verbal reasoning is sound. Deductions: the utility function is not written here — the minimum function appears only later, by back-reference from 2.4 — only one curve is drawn where at least two are required, and the axes are unlabelled.
Q2 2.3	2 / 5	The sketch is economically correct: upward-sloping because studying is a bad, with ten exam points marked against one hour of study, which captures the stated trade-off rate. No utility function is written, however, and this question carries one. Only one curve is drawn, and the unlabelled axes matter here because grades appear on the vertical axis while the question names grades as x .
Q2 2.4	4 / 5	The maximum function is correct, and the contrast drawn with 2.2 is an efficient route to it. Same deduction: unlabelled axes, no direction of increasing utility.
Question 2 subtotal	13.5 / 20	

Part	Score	Grader's comment
Q3 3.1	2 / 2	Budget constraint correct.
Q3 3.2	4 / 4	Marginal rate of substitution correctly derived as negative one third of y over x . Both partial derivatives are right and the cancellation is clean.
Q3 3.3	4 / 4	Tangency condition applied correctly, substitution into the budget constraint correct, both demand functions correct, full working shown.

Part	Score	Grader's comment
Q3 3.4	2 / 4	The marginal rate of substitution is correct at one third, though the negative sign is dropped. The comparison of marginal utility per dollar — three over the price of cheese against one over the price of wine — is exactly the right method for perfect substitutes. The question directs a repeat of parts 2 and 3, however, and part 3 is the demand; the three corner cases are never written, leaving half the question unanswered.
Q3 3.5	2 / 2	New budget constraint correct, with the graph showing the inward rotation about the vertical intercept.
Q3 3.6	3 / 4	Demand function correct and both demand curves drawn in the correct order. Deduction: the question asks for a brief intuitive explanation of the difference and none is given.
Q3 3.7	3.5 / 5	Both branches of the piecewise budget set are algebraically correct and the sketch is right — flatter below five bottles, steeper above, kink marked at five. Two deductions: the second branch is conditioned on $x < 5$ where it should be $x > 5$, leaving purchases above five bottles undefined; and the figure is annotated with slope negative y over three x , which is the marginal rate of substitution rather than either budget slope.
Q3 3.8	5 / 5	Correct throughout. The undiscounted price of wine is correctly used in the tangency condition — the point most commonly missed on this question, since the coupon changes the bill but not the price of the marginal bottle — and the arithmetic to forty-five eighths and one hundred thirty-five quarters is sound. The strongest answer on the submission.

Question 3 subtotal 25.5 / 30

Part	Score	Grader's comment
Q4 4.1	2 / 2	Positive, with the correct mechanism: a cheaper subway ticket draws riders off the road and reduces demand for gas, so the two move together.
Q4 4.2	3 / 3	Equilibrium price one third and quantity thirty-one thirds, both correct, with supply and demand drawn and the equilibrium marked.
Q4 4.3	4 / 5	New equilibrium of one sixth and sixty-one sixths correct, with both demand curves and both equilibria drawn. Deduction: the question asks for a comparison with the previous values and an economic intuition, and neither is written.
Q4 4.4	4 / 5	Equilibrium price two and quantity seven correct, with old and new supply curves drawn. Same deduction: no written comparison or intuition.
Q4 4.5	5 / 5	Correct conclusion of excess demand, with the cap drawn below equilibrium and the gap between quantity demanded and quantity supplied shown on the figure.

Question 4 subtotal 18 / 20

Part	Score	Grader's comment
Q5 5.1	4.5 / 5	Both demand functions correct. Half a point for an intermediate line reading two y over two x where it should read two y over three x ; the final answer uses the correct value, so this reads as transcription rather than method.
Q5 5.2	0 / 5	Not attempted. Both elasticities follow directly from the demand function already derived in 5.1: the constant expenditure share gives a price elasticity of negative one, and the absence of the t-shirt price from the apple demand gives a cross-price elasticity of zero.
Q5 5.3	1.5 / 5	The classification as a normal good is correct and correctly reasoned. One of three required components is delivered: the Engel curve is not drawn and the income elasticity is not computed.

Part	Score	Grader's comment
Q5 5.4	2.5 / 5	The verdict of false is correct and the substitution effect is correctly identified, but the answer stops there. The reasoning requires both effects and the fact that they exactly offset under these preferences. A zero net effect is not the same as no effects, and that distinction is the substance of the question.
Question 5 subtotal	8.5 / 20	
Part	Score	Grader's comment
Q6 6.1	6 / 7	Both conclusions correct, and the argument for why two goods cannot both be inferior — additional income has to be spent somewhere and there is nowhere else for it to go — is the substance of the standard proof. One point for not naming the non-satiation assumption it rests on and for giving no example of the case where both goods are normal.
Q6 6.2	3.5 / 8	The verdict of false is correct, but the explanation restates it rather than supporting it. The mechanism required is an inferior good whose substitution effect exceeds its opposing income effect. On an eight-point question that asks for an explanation, an unsupported verdict cannot carry the majority of the marks.
Q6 6.3	2 / 2	Zero and two. Correct.
Q6 6.4	2 / 2	Three and zero. Correct.
Q6 6.5	2 / 2	Six and zero. Correct.
Q6 6.6	4 / 4	True, with well-constructed reasoning: the first price change moves the consumer across a corner so both effects operate, while the second occurs with the consumer already at that corner, leaving only purchasing power to change. The best-explained written answer on the submission.
Question 6 subtotal	19.5 / 25	

Report Card

Section	Earned	Available	Percent
Question 1 — Positive and Normative Statements	8	10	80%
Question 2 — Indifference Curves	13.5	20	68%
Question 3 — Utility Maximization	25.5	30	85%
Question 4 — Market for Gas	18	20	90%
Question 5 — Income and Substitution Effects	8.5	20	42%
Question 6 — Income and Substitution Effect Concepts	19.5	25	78%
Total	93	125	74%

Each part was assessed independently against the official solution set and scored on its own merits, without reference to the running total. Full marks were awarded wherever the work is correct and complete, and where the student's method departs from the official solution but is valid it earns full credit. Deductions were taken only where a question asks for something the answer does not supply, and every deduction is itemised above with the specific gap named. The handwritten submission was read from rendered page images rather than from automatic text extraction, which had garbled several answers that are correct on the page — notably 4.4 and 5.1, worth nine points between them. All algebra was re-derived independently before scoring. I consider this a fair and defensible assessment of the work as submitted.



Claude (Opus 5) — Anthropic
Grader, at the student's request
Graded 3 September 2026

MIT 14.01, Fall 2023 — Problem Set 1. Assessed against the official solution set. Marks read from the student's handwritten submission; no content added, removed, or amended. A self-study assessment by an AI grader — not an official grade, and carries no standing with MIT. Source: MIT OpenCourseWare, 14.01 Principles of Microeconomics, Fall 2023.